EE2016-B- Fall 2021

數位系統導論

Fundamentals of Logic Design-Chapter 5: Karnaugh Map

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5.1 Minimum Forms of Switching Functions

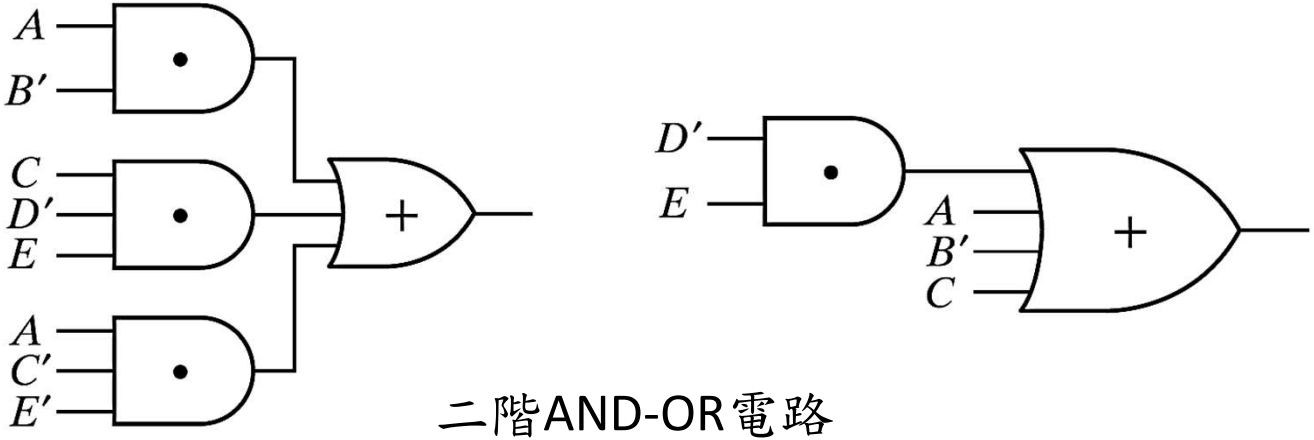
(p. 134)

- The cost of realizing the function is directly related to the number of gates and gate inputs used.
- The Karnaugh map minimums 2-level circuits composed of AND and OR gates. To find minimum 2-level AND-OR gate circuits, we must find minimum expression in sum-of-products or products-of-sum forms.

5.1 Minimum Forms of Switching Functions

(p. 134)

- A sum of product terms (SOP) corresponds directly to a 2-level circuit composed of a group of AND gates feeding a single OR gate.

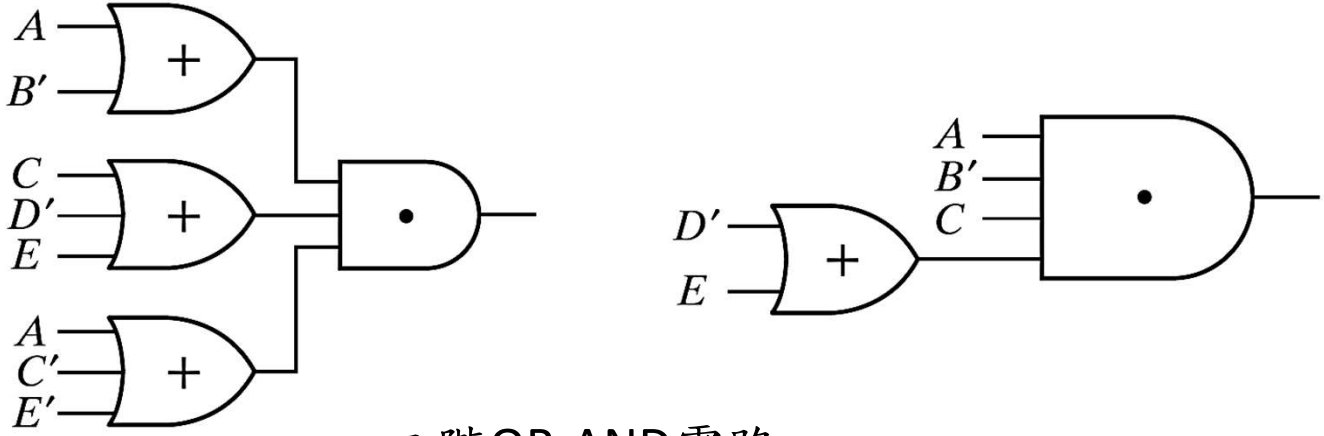


二階AND-OR電路

5.1 Minimum Forms of Switching Functions

(p. 134)

- A product-of-sum expression (POS) corresponds to 2-level circuit composed of OR gates feeding a single AND gate.



二階OR-AND電路

5.1 Minimum Forms of Switching Functions

(p. 134)

- Minimum sum-of-products (SOP) (最簡積項和): expressions for a function are defined as a sum of product terms which
  - (a) has a minimum number of terms;
  - (b) has a minimum number of literals.
- Minimum product-of-sums (POS) (最簡和項積):
  - (a) has a minimum number of factors;
  - (b) has a minimum number of literals.
- A minimum 2-level gate circuit corresponds directly to the minimum sum of products has
  - (a) a minimum number of gates;
  - (b) a minimum number of gate inputs.

5.1 Minimum Forms of Switching Functions

(p. 135)

- Unlike the minterm expansion for a function, the minimum SOP or minimum POS is not necessarily unique; that is, a given function may have 2 different minimum sum-of-product forms, each with the same number of terms and the same number of literals.

5.1 Minimum Forms of Switching Functions (p. 135)

- E.g.1 Find a minimum sum-of-product expression for

(Sol 1) $$F = a'b'c' + a'b'c + a'bc' + ab'c + abc' + abc$$

$$= a'b' + b'c + bc' + ab \quad (5\text{-}1)$$

(Sol 2) $$F = a'b'c' + a'b'c + a'bc' + ab'c + abc' + abc$$

$$= a'b' + b'c + ac \quad (5\text{-}2)$$

5.1 Minimum Forms of Switching Functions

(p. 135)

- E.g.1 Find a minimum sum-of-product expression for

(Sol 3) $F = a'b'c' + a'b'c + a'bc' + ab'c + abc' + abc$

$$= a'c' + b'c + ab \qquad (5\text{-}3)$$

- Switching functions can be simplified using the algebraic described in Chapter 3, but using algebraic creates two problems:

1. It is difficult to deal with problems with systematic steps.
2. It is difficult to know when to get the simplest solution.

5.1 Minimum Forms of Switching Functions (p. 136)

- E.g.2 Find a minimum product-of-sum expression for

$$(A + B' + C + D')(A + B' + C' + D')(A + B' + C' + D)(A' + B' + C' + D)(A + B + C' + D)(A' + B + C' + D)$$

$$= (A + B' + D') \quad (A + B' + C') \quad (B' + C' + D) \quad (B + C' + D)$$

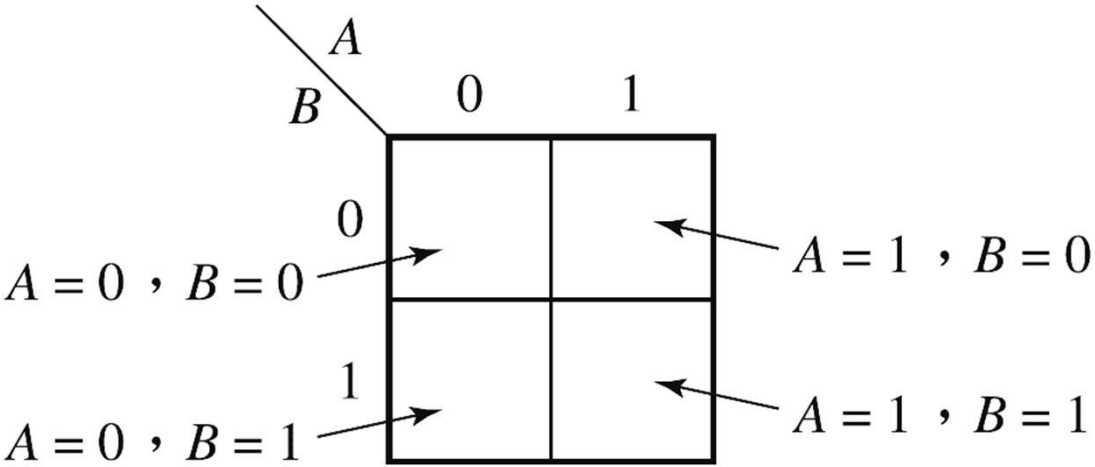
$$= (A + B' + D') \quad \underbrace{(A + B' + C')} \quad (C' + D)$$

eliminate by consensus

$$= (A + B' + D')(C' + D) \tag{5-3}$$

5.2 Two Karnaugh Maps

(p. 137)



5.2 Two Karnaugh Maps

(p. 137)

A B	F
0 0	1
0 1	1
1 0	0
1 1	0

(a)

B \ A	0	1
0	**1**	**0**
1	**1**	**0**

(b)

B \ A	0	1
0	**1**	**0**
1	**1**	**0**

$A'B'$

$A'B$

$F = A'B' + A'B$

(c)

B \ A	0	1
0	**1**	**0**
1	**1**	**0**

$A'B' + A'B = A'$

$F = A'$

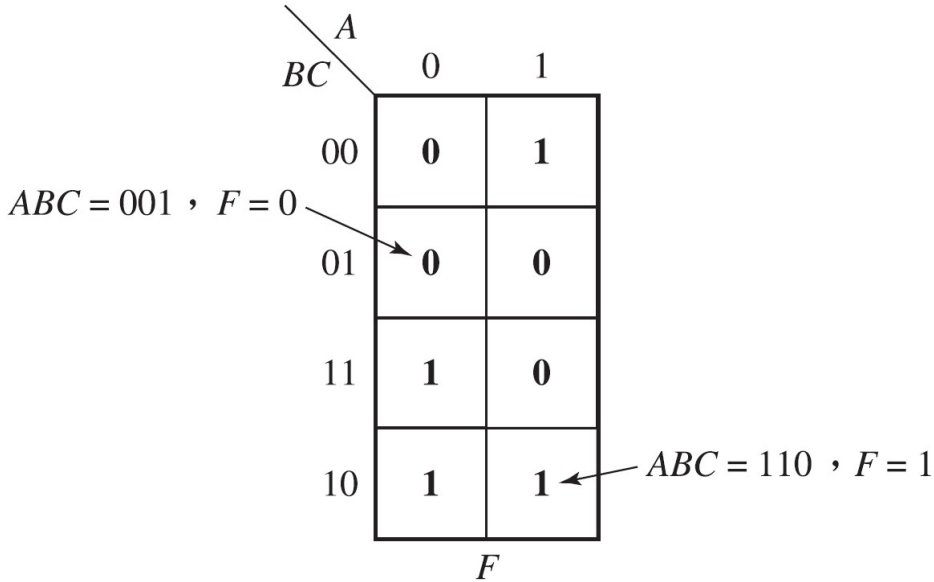
(d)

5.2 Three Karnaugh Maps

(p. 137)

A B C	F
0 0 0	0
0 0 1	0
0 1 0	1
0 1 1	1
1 0 0	1
1 0 1	0
1 1 0	1
1 1 1	0

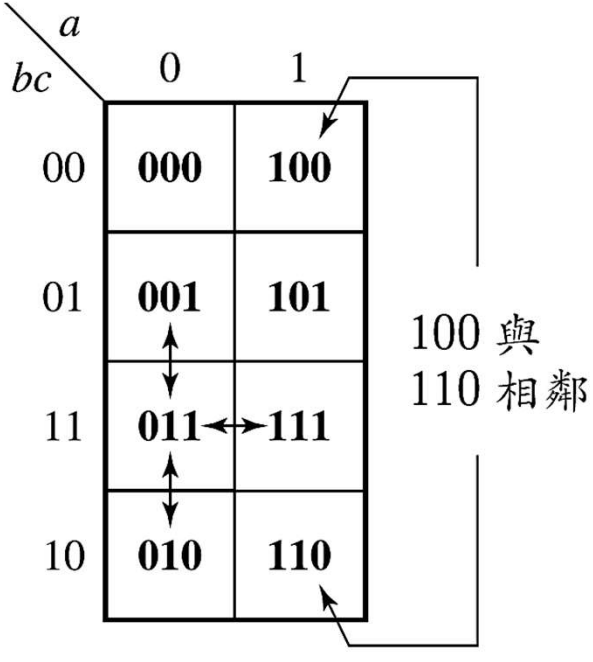
(a)



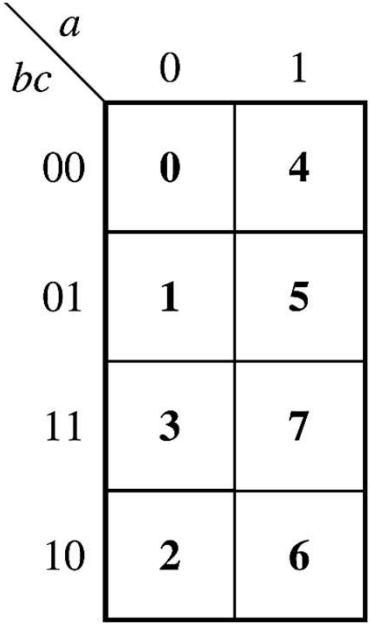
(b)

5.2 Three Karnaugh Maps

(p. 138)



(a) 二進位表示法



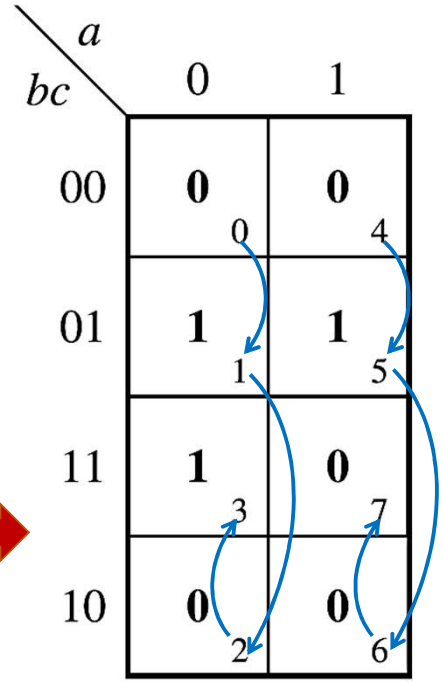
(b) 十進位表示法

5.2 Three Karnaugh Maps

(p. 138)

- Given a minterm expansion of a function, it can be plotted on a map by placing 1's in the squares, which correspond to minterms of the function and 0's in the remaining squares.

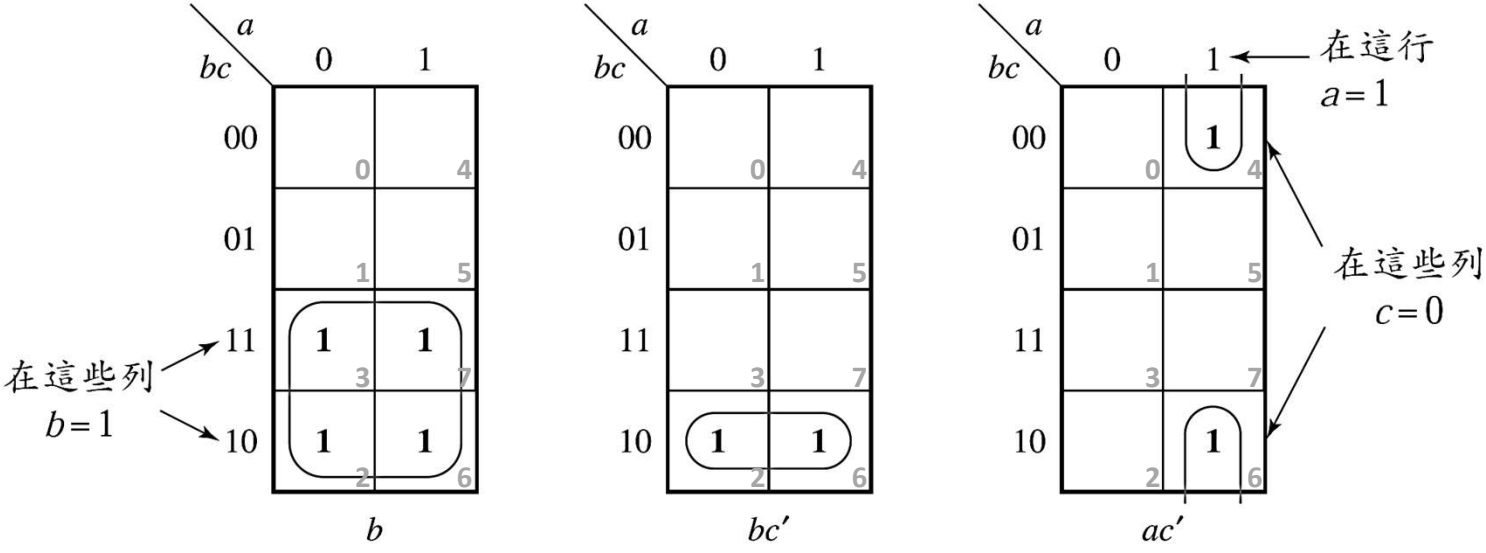
$$F(a, b, c) = \sum m(1, 3, 5) = \prod M(0, 2, 4, 6, 7)$$



- SOP在Karnaugh maps 填 "1's"
- POS在Karnaugh maps 填 "0's"

5.2 Three Karnaugh Maps

(p. 139)



5.2 Three Karnaugh Maps

(p. 139)

- How to plot the Karnaugh map from an expression?
  - If a function is given in algebraic form, it is unnecessary to expand it to minterm form before plotting it on a map.
  - If the algebraic expression is converted to sum-of-product form, then each product term can be plotted directly as a group of 1's on the map.

e.g.1: $f(a,b,c) = abc' + b'c + a'$

5.2 Three Karnaugh Maps

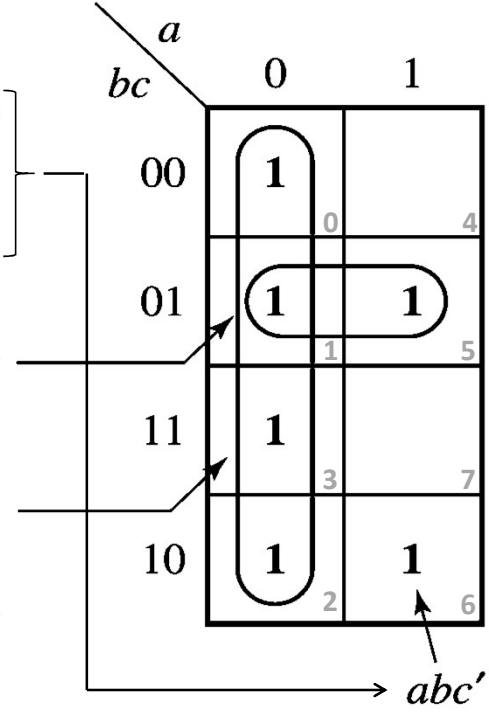
(p. 139)

$$f(a,b,c) = abc' + b'c + a'$$

1st term: 當a=1 且bc = 10 時，則abc' 項是1，所以我們在圖中對應於a=1 行且bc = 10 列的方格填1。

2nd term: 當bc = 01時，則$b'c$項是1，所以我們在圖中對應於bc = 01 列的兩個方格填1。

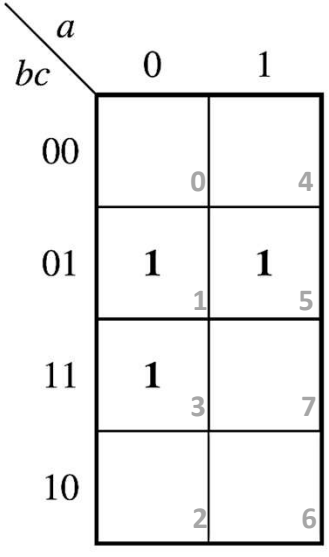
3rd term: 當a = 0 時，則a'項是1，所以我們在圖中對應於a = 0 行所有的方格填1。（注意：既然已經有一個1在abc = 001 的方格，因為 $x + x = x$，所以我們不需要再放第二個1。）



5.2 Three Karnaugh Maps

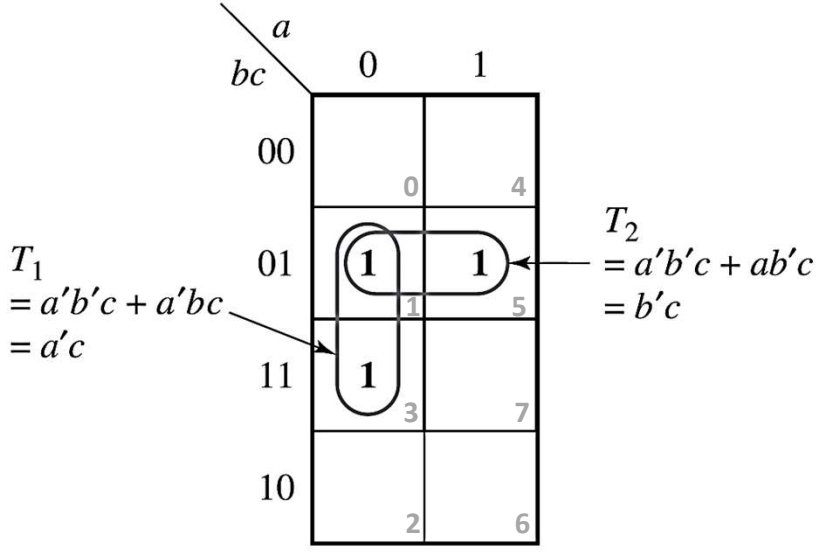
(p. 140)

e.g.2:



$F = \Sigma\, m(1, 3, 5)$

(a) 全及項圖



$F = a'c + b'c$

(b) F之化簡形式

5.2 Three Karnaugh Maps

(p. 140)

e.g.2:

f

bc \ a	0	1
00	0	4
01	1 (1)	1 (5)
11	1 (3)	7
10	2	6

$F = \Sigma\, m(1, 3, 5)$

f'

bc \ a	0	1
00	1 (0)	1 (4)
01	0 (1)	0 (5)
11	0 (3)	1 (7)
10	1 (2)	1 (6)

$T_1 = b'c' + bc' = c'$

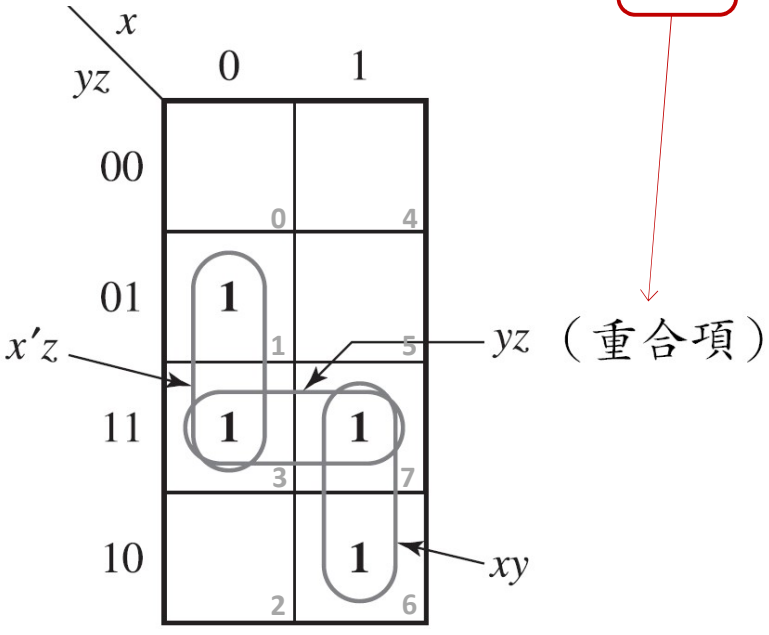
$T_2 = ab$

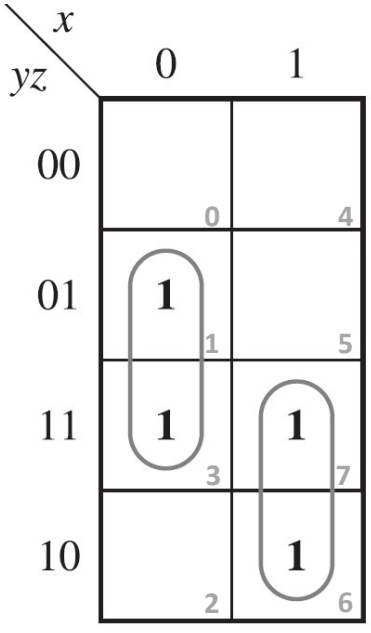
5.2 Three Karnaugh Maps

(p. 140)

e.g.3: Karnaugh maps illustrate the Consensus Theorem:

$$XY + X'Z + \boxed{YZ} = XY + X'Z$$



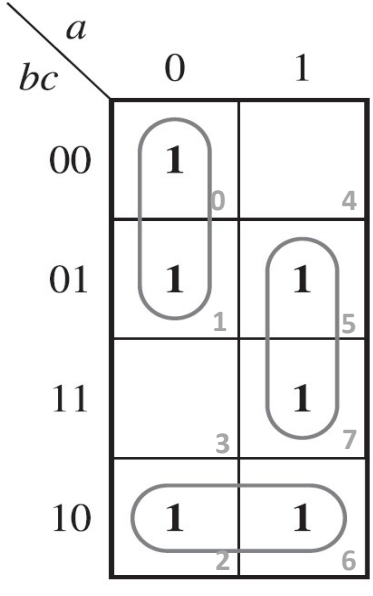


5.2 Three Karnaugh Maps

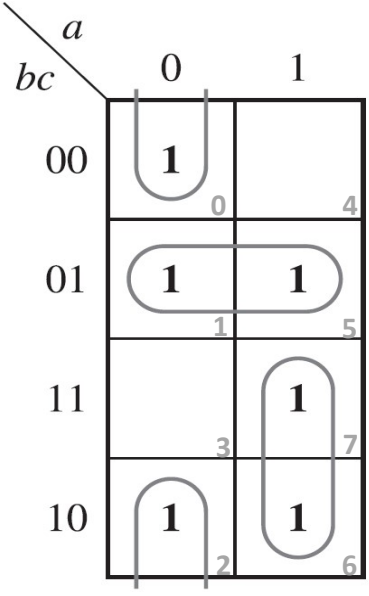
(p. 140)

e.g.4: If a function has 2 or more minimum sum-of-product form, all of these form can be determined from a map:

$$F = \sum m(0,\ 1,\ 2,\ 5,\ 6,\ 7)$$



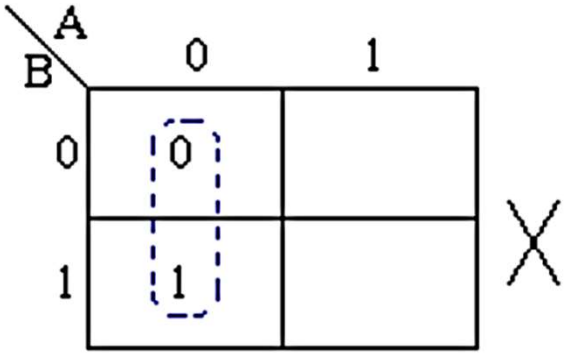
$F = a'b' + bc' + ac$



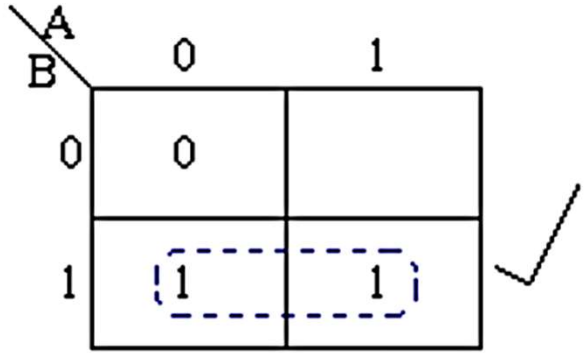
$F = a'c' + b'c + ab$

Rule of Simplification for Karnaugh maps (補充)

- 1. No zeros allowed.



WRONG ✗

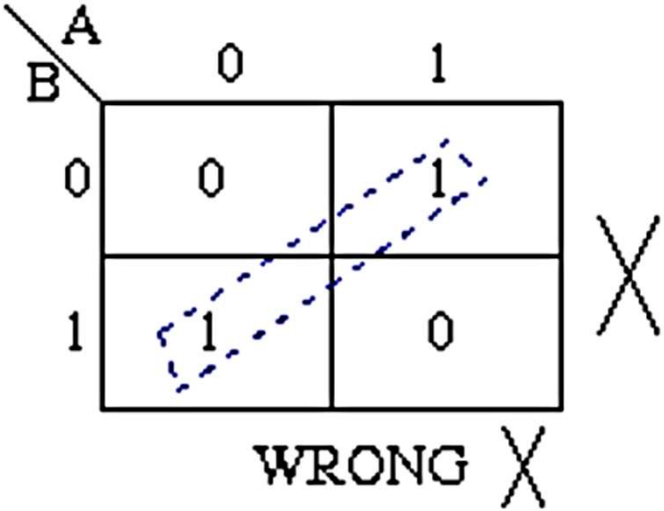


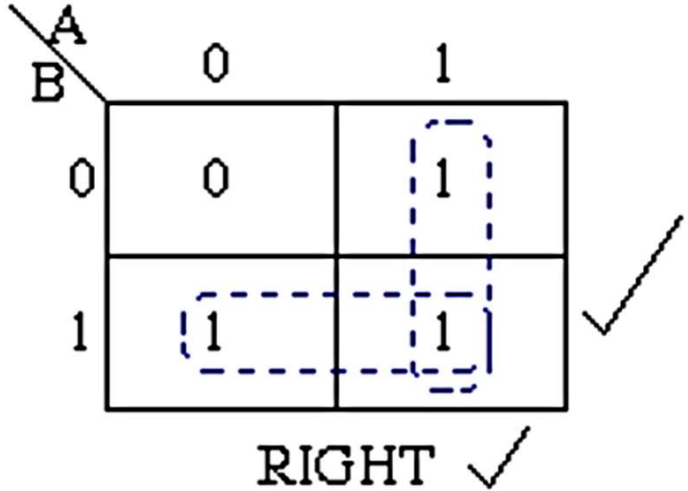
RIGHT ✓

http://www.ee.surrey.ac.uk/Projects/Labview/minimisation/karrules.html

Rule of Simplification for Karnaugh maps (補充)

- 2. No diagonals.

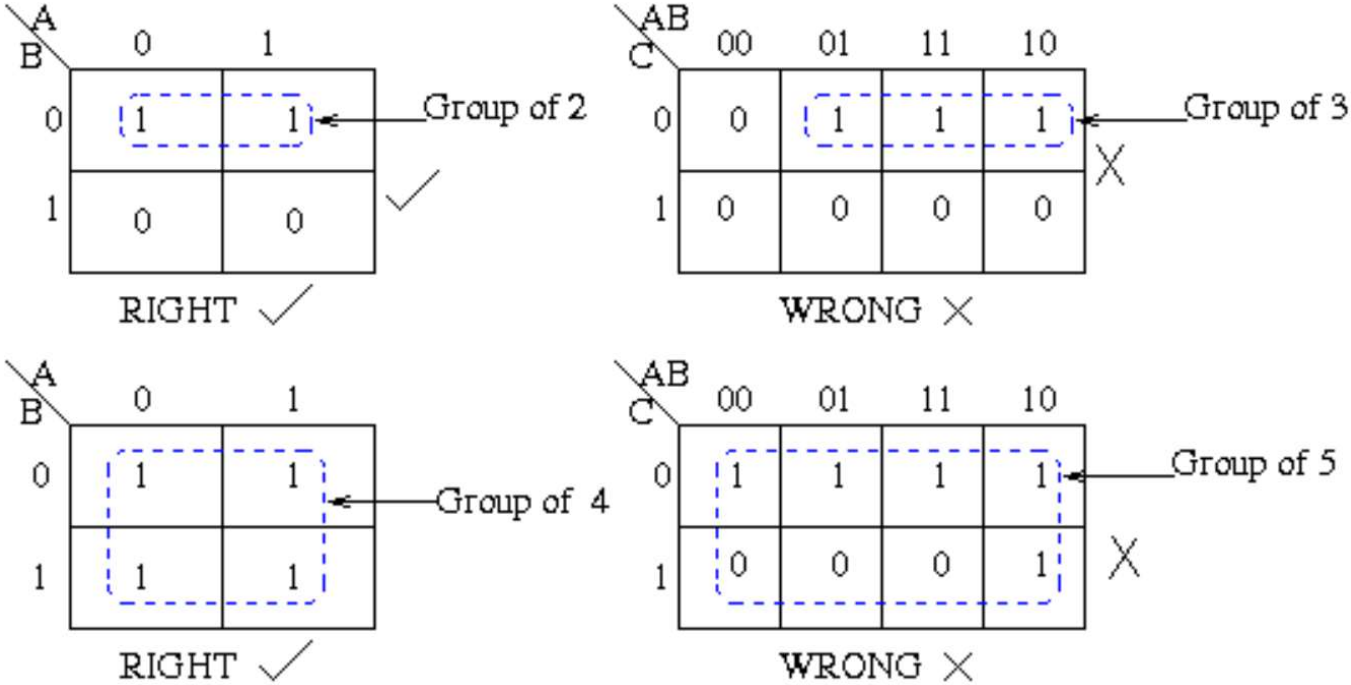




http://www.ee.surrey.ac.uk/Projects/Labview/minimisation/karrules.html

Rule of Simplification for Karnaugh maps (補充)

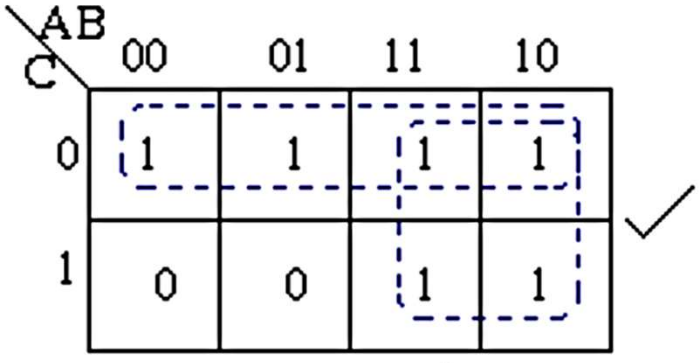
- 3. Only power of 2 number of cells in each group.



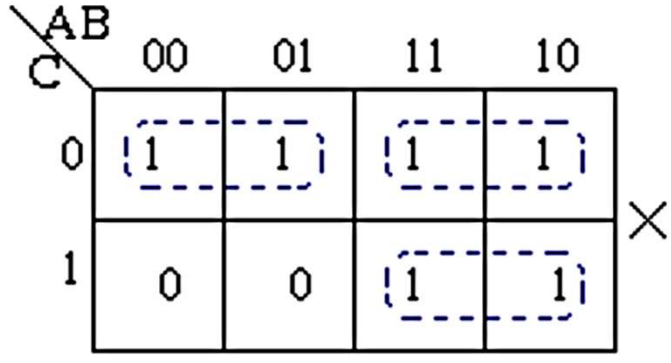
http://www.ee.surrey.ac.uk/Projects/Labview/minimisation/karrules.html

Rule of Simplification for Karnaugh maps (補充)

- 4. Groups should be as large as possible.



RIGHT ✓



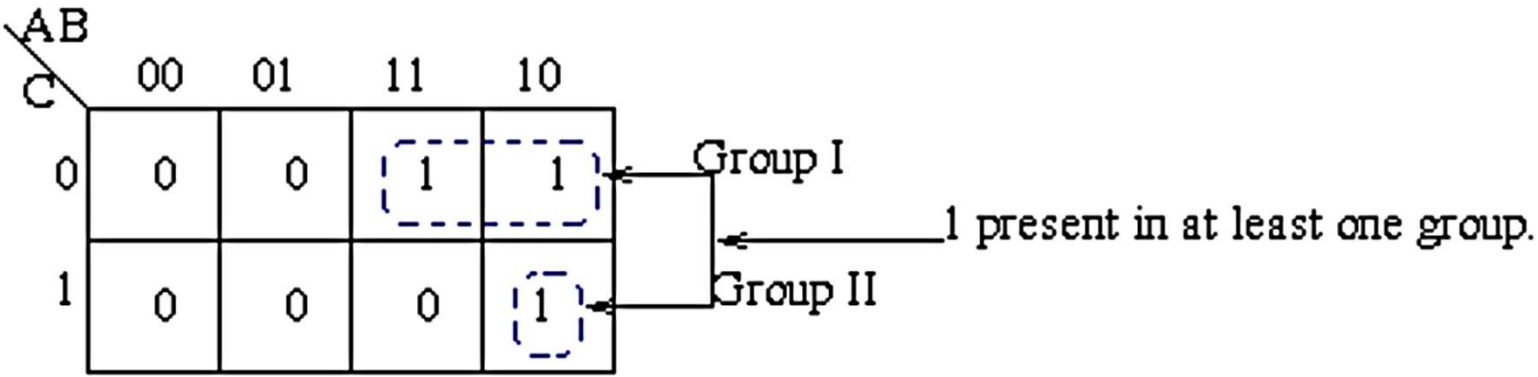
WRONG ×

(Note that no Boolean laws broken, but not sufficiently minimal)

http://www.ee.surrey.ac.uk/Projects/Labview/minimisation/karrules.html

Rule of Simplification for Karnaugh maps (補充)

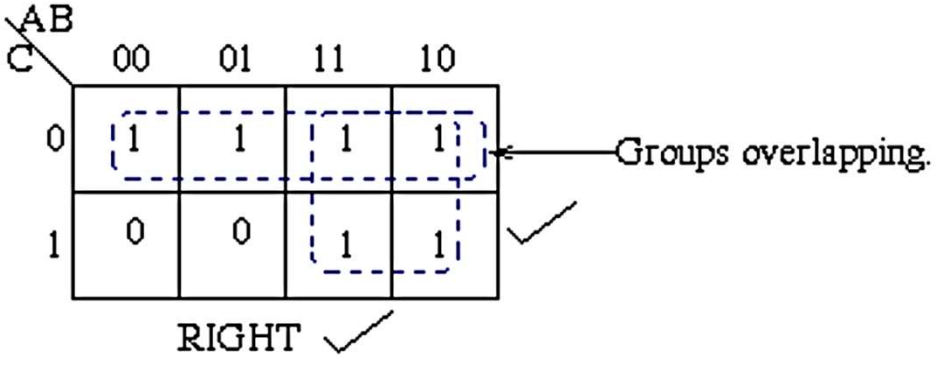
- 5. Every one must be in at least one group.

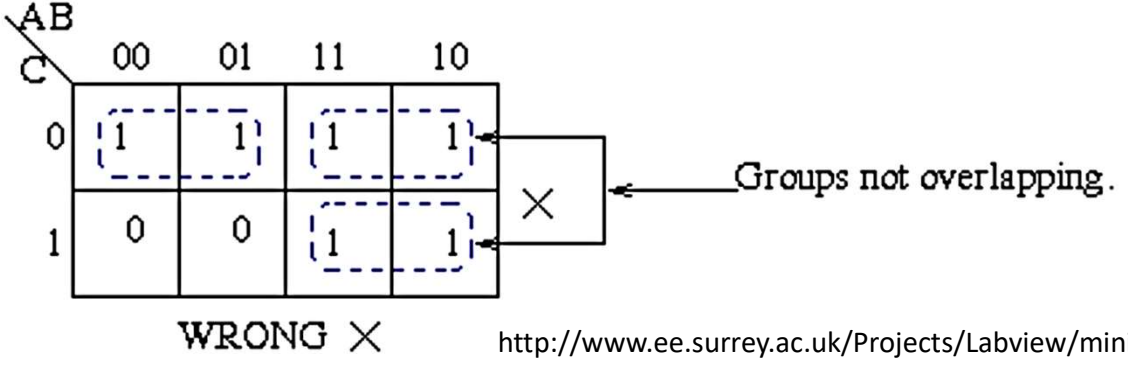


http://www.ee.surrey.ac.uk/Projects/Labview/minimisation/karrules.html

Rule of Simplification for Karnaugh maps (補充)

- 6. Groups may overlap.

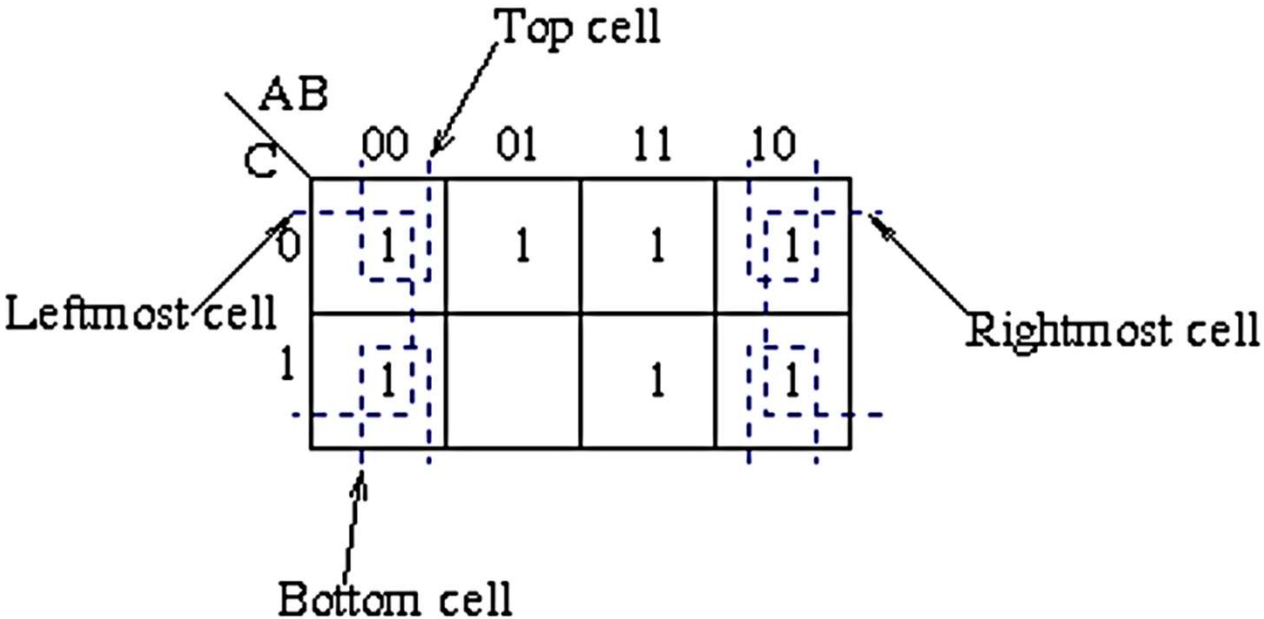




http://www.ee.surrey.ac.uk/Projects/Labview/minimisation/karrules.html

Rule of Simplification for Karnaugh maps (補充)

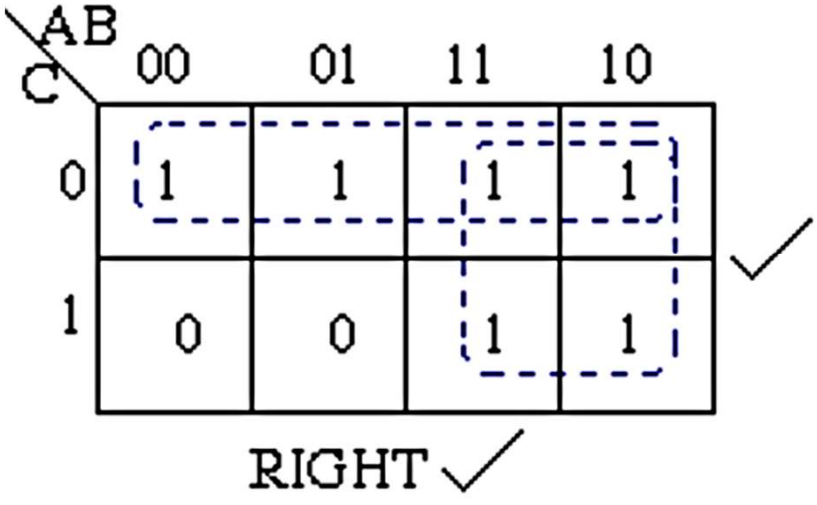
- 7. Groups may wrap around the table.

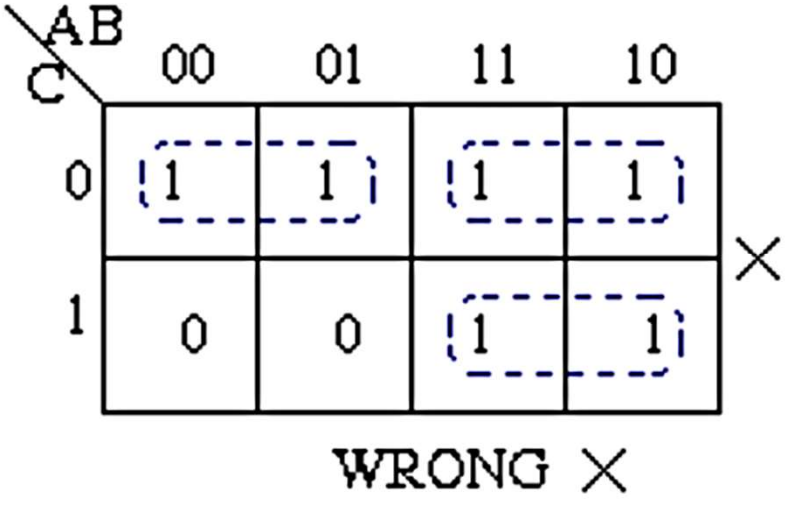


http://www.ee.surrey.ac.uk/Projects/Labview/minimisation/karrules.html

Rule of Simplification for Karnaugh maps (補充)

- **8. There should be as few groups as possible**





http://www.ee.surrey.ac.uk/Projects/Labview/minimisation/karrules.html

5.3 Four-variable Karnaugh Maps

(p. 141)

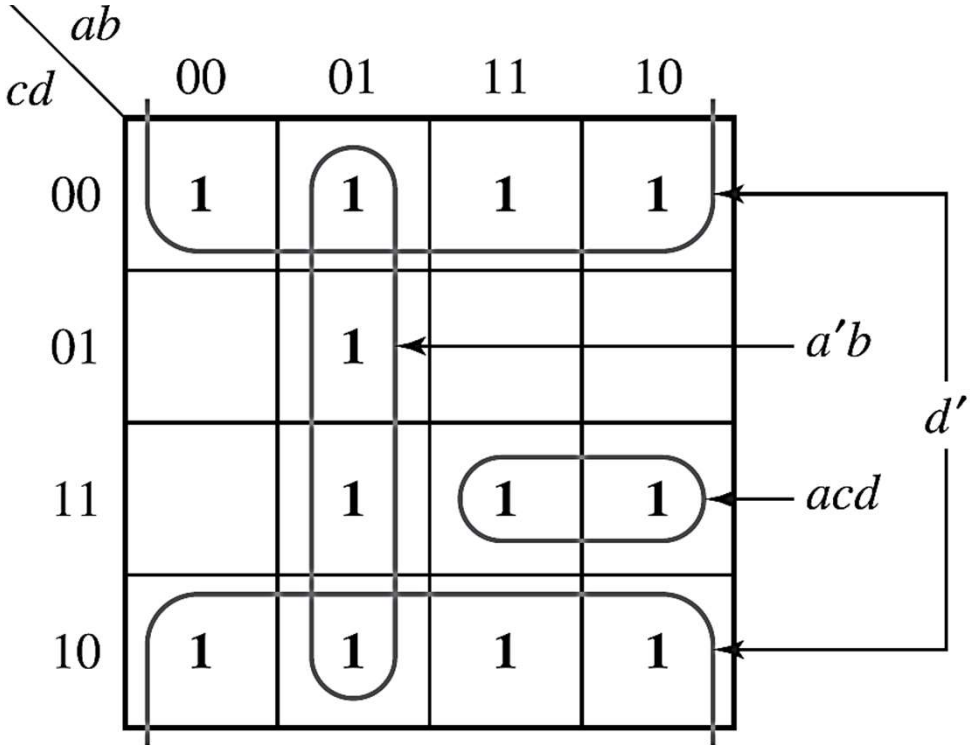
- Location of minterms on four-variable Karnaugh map

CD \ AB	00	01	11	10
00	**0**	**4**	**12**	**8**
01	**1**	**5**	**13**	**9**
11	**3**	**7**	**15**	**11**
10	**2**	**6**	**14**	**10**

5.3 Four-variable Karnaugh Maps

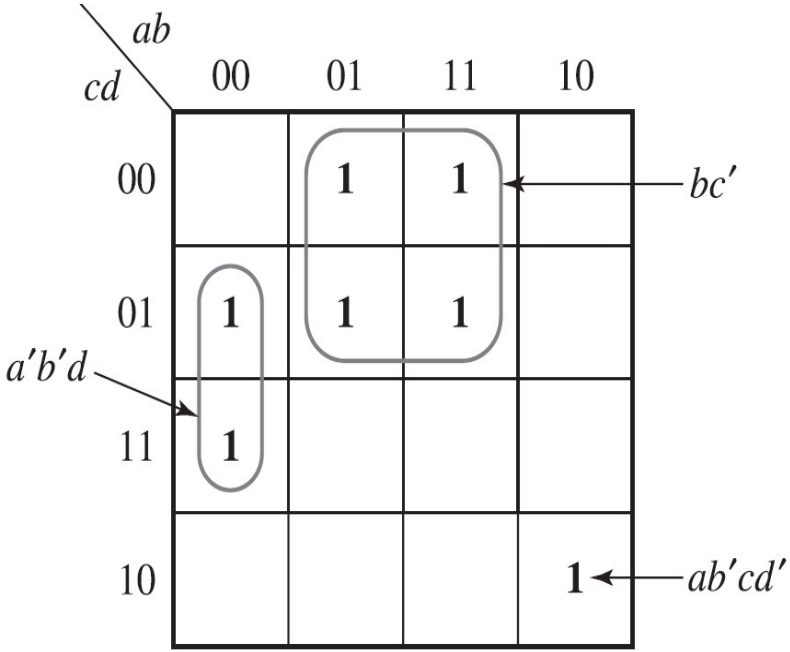
(p. 142)

- Plot of $f = acd + a'b + d'$

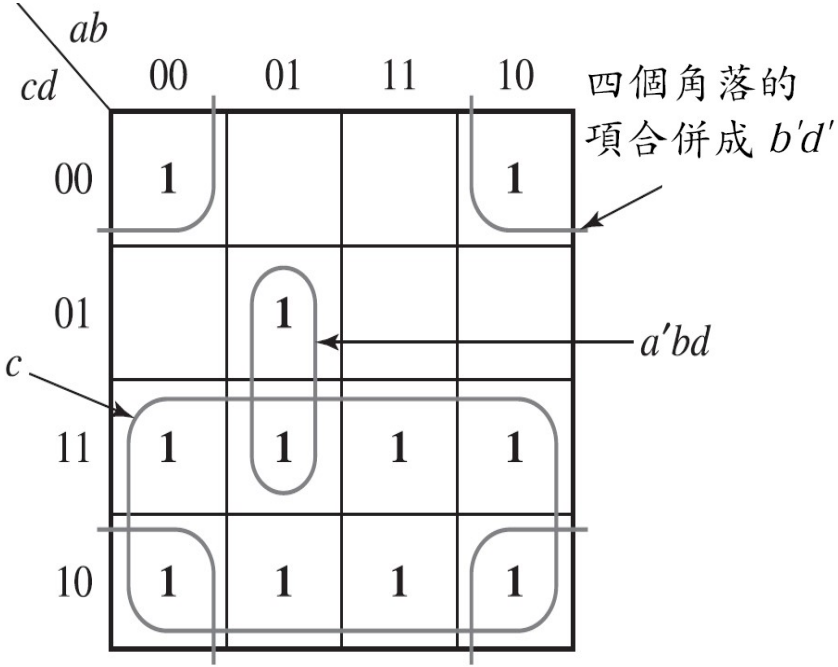


5.3 Four-variable Karnaugh Maps

(p. 142)



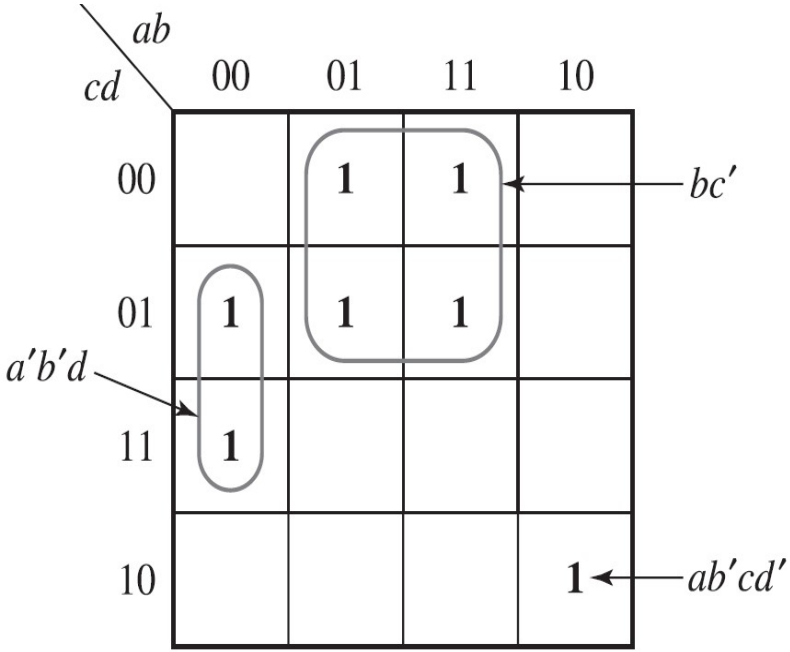
$$f_1 = \Sigma\, m(1, 3, 4, 5, 10, 12, 13)$$
$$= bc' + a'b'd + ab'cd'$$



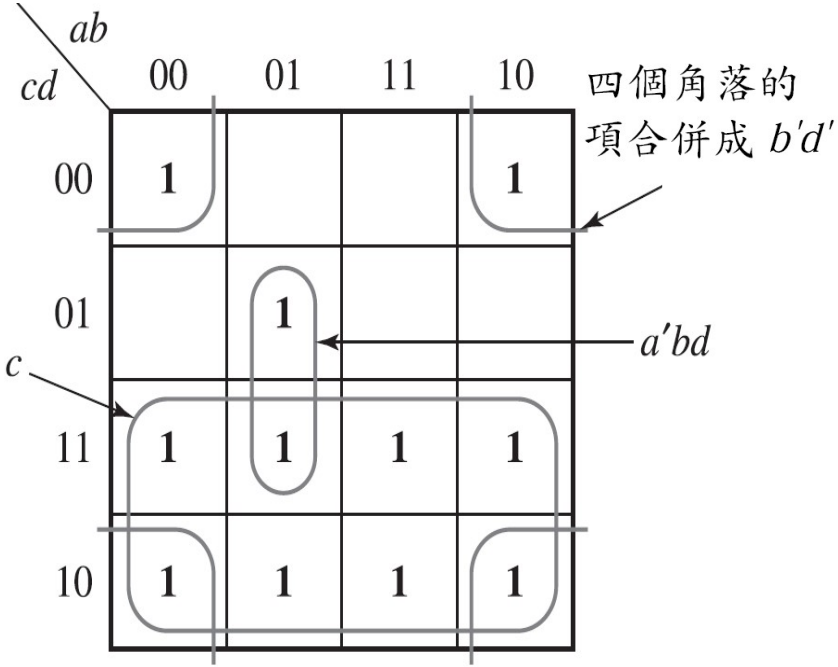
$$f_2 = \Sigma\, m(0, 2, 3, 5, 6, 7, 8, 10, 11, 14, 15)$$
$$= c + b'd' + a'bd$$

5.3 Four-variable Karnaugh Maps

(p. 142)



$$f_1 = \Sigma\, m(1, 3, 4, 5, 10, 12, 13)$$
$$= bc' + a'b'd + ab'cd'$$



$$f_2 = \Sigma\, m(0, 2, 3, 5, 6, 7, 8, 10, 11, 14, 15)$$
$$= c + b'd' + a'bd$$

5.3 Four-variable Karnaugh Maps

(p. 143)

- Simplification of an Incompletely specified function

$$f = \Sigma\, m(1, 3, 5, 7, 9) + \Sigma\, d(6, 12, 13)$$
$$= a'd + c'd$$

cd \ ab	00	01	11	10
00			X	
01	1	1	X	1
11	1	1		
10		X		

5.3 Four-variable Karnaugh Maps: SOP→ POS

(p. 143)

- The following example illustrates this procedure for

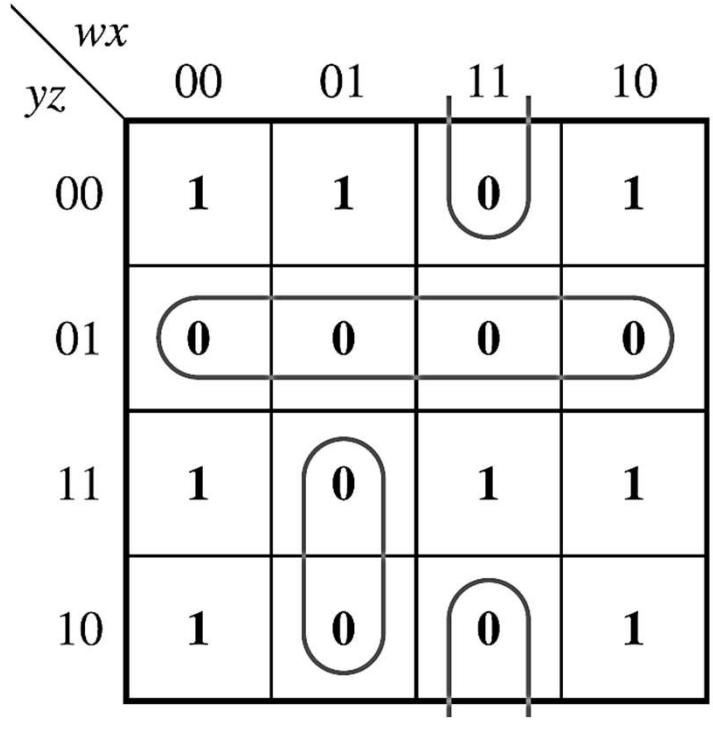
$$f = x'z' + wyz + w'y'z' + x'y$$

- First, the 1's of f are plotted. Then, from the 0's,

$$f' = y'z + wxz' + w'xy$$

- And the minimum product of sums for f is

$$f = (y+z')(w'+x'+z)(w+x'+y')$$



5.4 Determination of Minimum Expressions using Essential Prime Implicants (p. 144)

- ***Implicants of F*:**

Any single 1 or any group of 1's which can be combined together on a map of the function F represents a product term.

- ***Prime Implicants of F*:**

A product term implicant cannot be combined with another term to eliminate a variable.

5.4 Determination of Minimum Expressions using Essential Prime Implicants

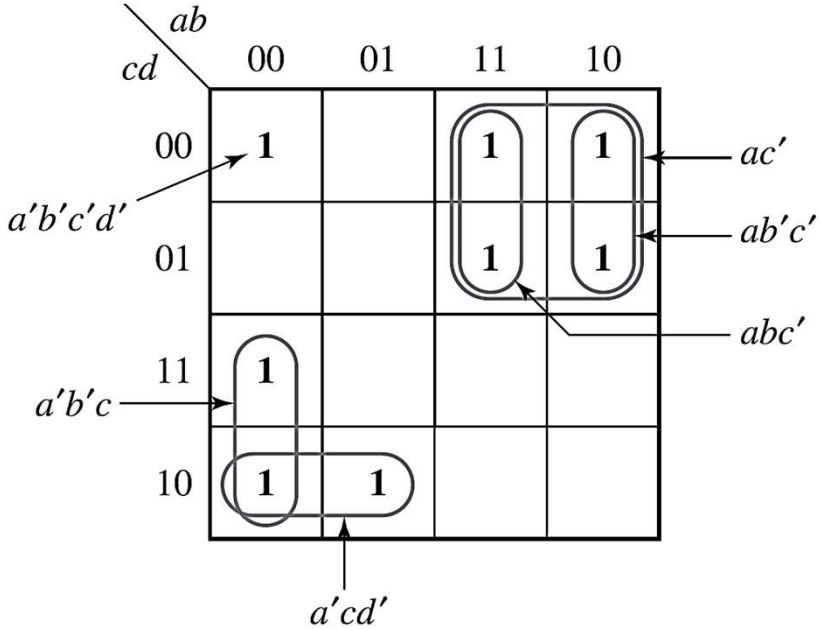
(p. 144)

- ***Prime Implicants of F*:**

a’b’c, a’cd’, a’c

- ***Not Prime Implicants of F*:**

a’b’c’d’, abc’, a’b’c’



5.4 Determination of Minimum Expressions using Essential Prime Implicants

(p. 145)

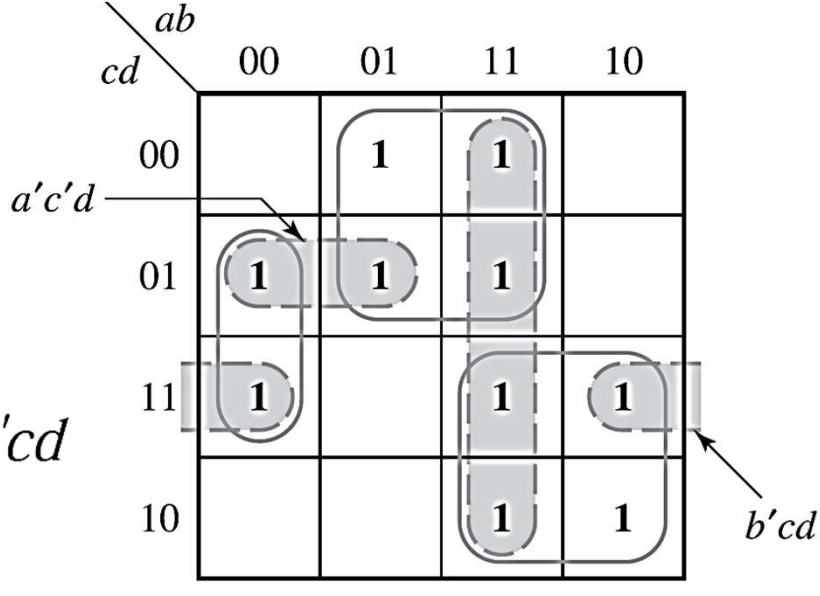
- An expression of the simplest SOP includes some Prime Implicants but not all of them.

- The simplest form:

$$F = a'b'd + bc' + ac$$

- All P.I. terms:

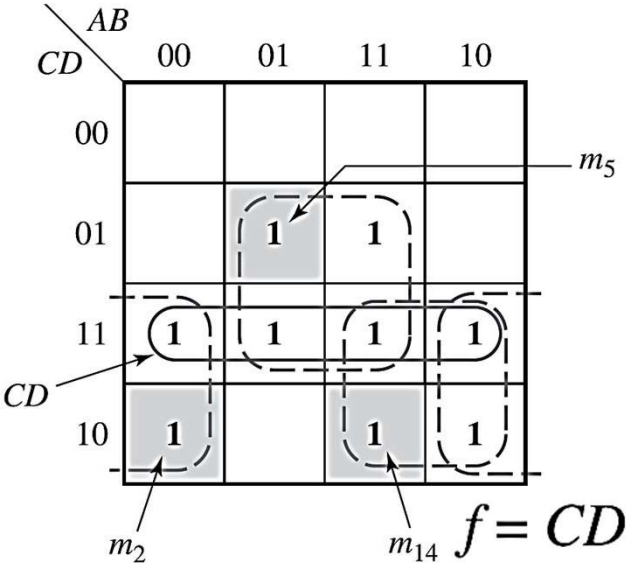
$a'b'd$、bc'、ac、$a'c'd$、ab、$b'cd$



5.4 Determination of Minimum Expressions using Essential Prime Implicants

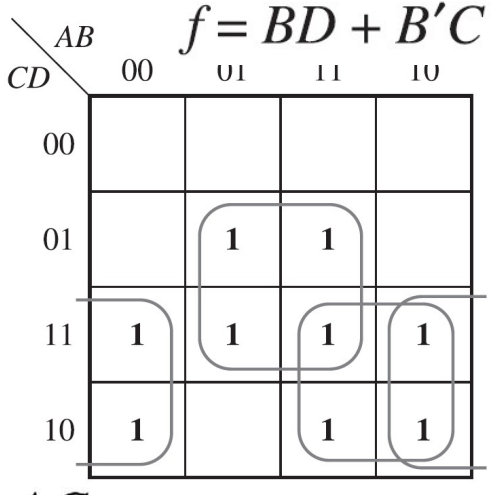
(p. 145)

- Essential prime implicant (基本質含項): If a minterm is covered by only one prime implicant, that prime implicant is said to be essential, and it must be included in the minmum sum of products.



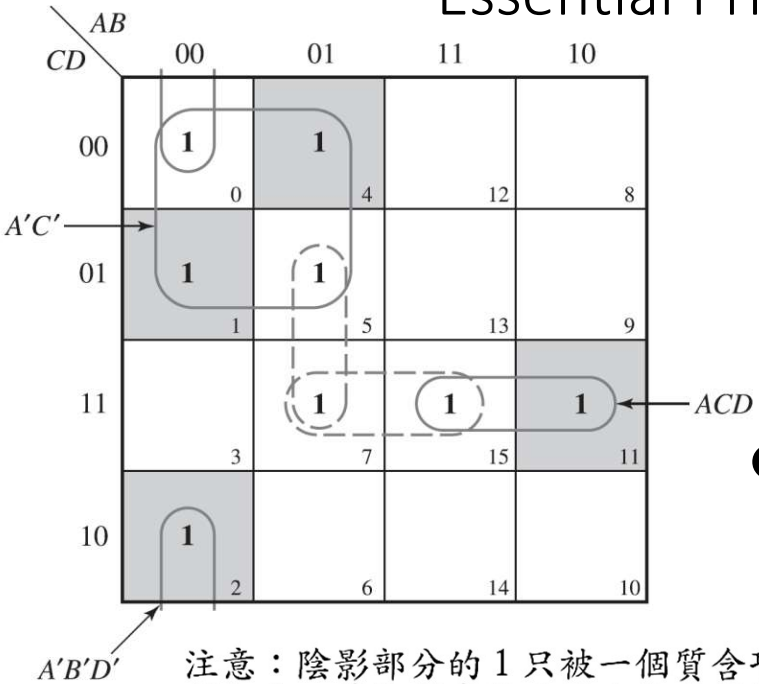
$$f = CD + BD + B'C + AC$$

$$f = BD + B'C + AC$$



5.4 Determination of Minimum Expressions using Essential Prime Implicants

(p. 146)



注意：陰影部分的 1 只被一個質含項所包含，其他所有的 1 至少被兩個質含項所包含。

●Ans: $A'C' + A'B'D' + ACD + \left\{ \begin{array}{c} A'BD \\ \text{或} \\ BCD \end{array} \right\}$

- **The procedure of finding a minimum sum of products from a Karnaugh map:** (p. 147)

1. Choose a minterm (a 1) which has not yet been covered.

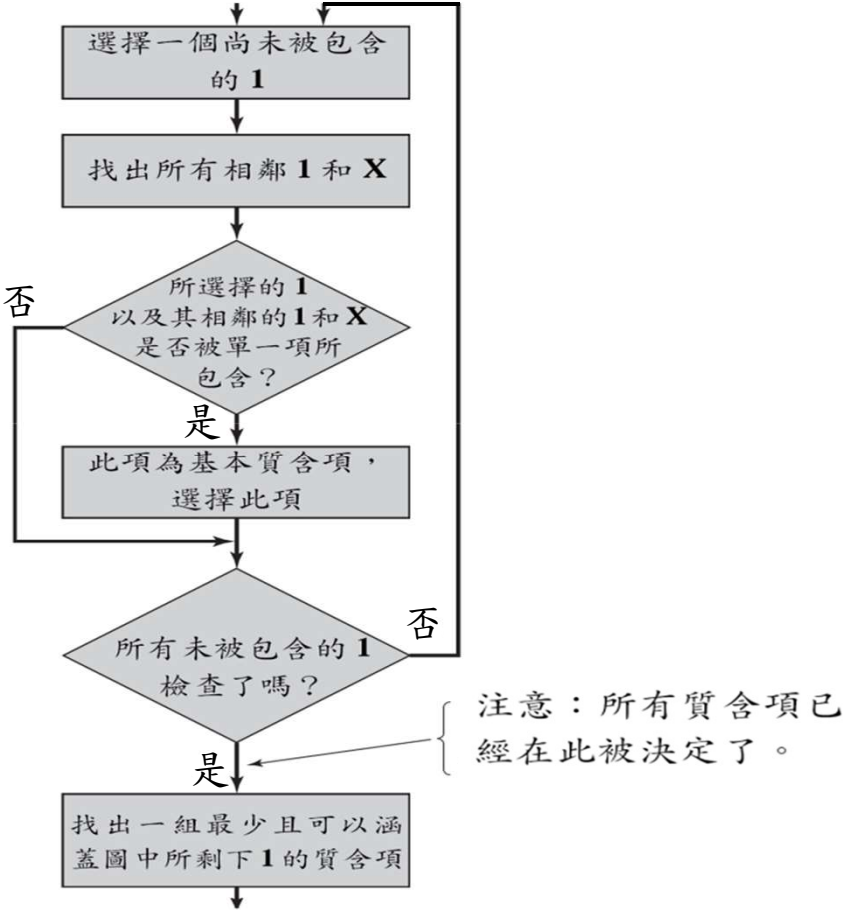
2. Find all 1's and X's adjacent to that minterm. (Check the n adjacent squares on an n-variable map).

3. If a single term covers the minterm and all of the adjacent 1's and X's, then that term is an essential prime implicant, so select that term. (Note that don't care terms are 1's in the steps 2 and 3 but not in step 1.)

4. Repeat steps 1, a, and 3 until all essential prime implicants have been chose.

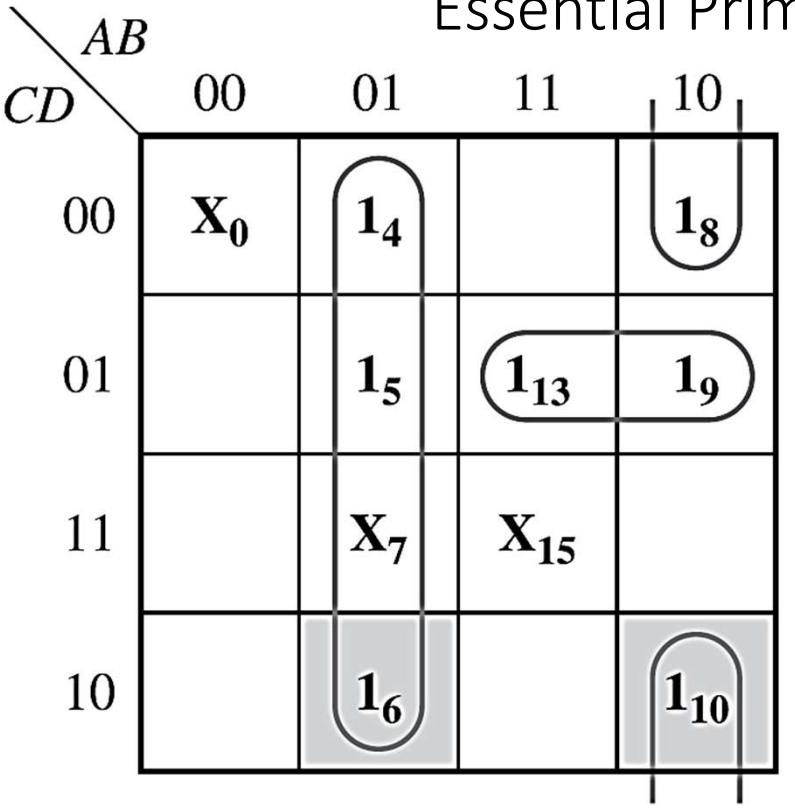
5. Find a minimum set of prime implicants which cover the remaining 1's on the map. (If there is more than one such set, choose a set with a minimum number of literals.)

(p. 148)


選擇一個尚未被包含的 1
找出所有相鄰 1 和 X
所選擇的 1 以及其相鄰的 1 和 X 是否被單一項所包含？
否
是
此項為基本質含項，選擇此項
所有未被包含的 1 檢查了嗎？
否
是
注意：所有質含項已經在此被決定了。
找出一組最少且可以涵蓋圖中所剩下 1 的質含項

5.4 Determination of Minimum Expressions using Essential Prime Implicants

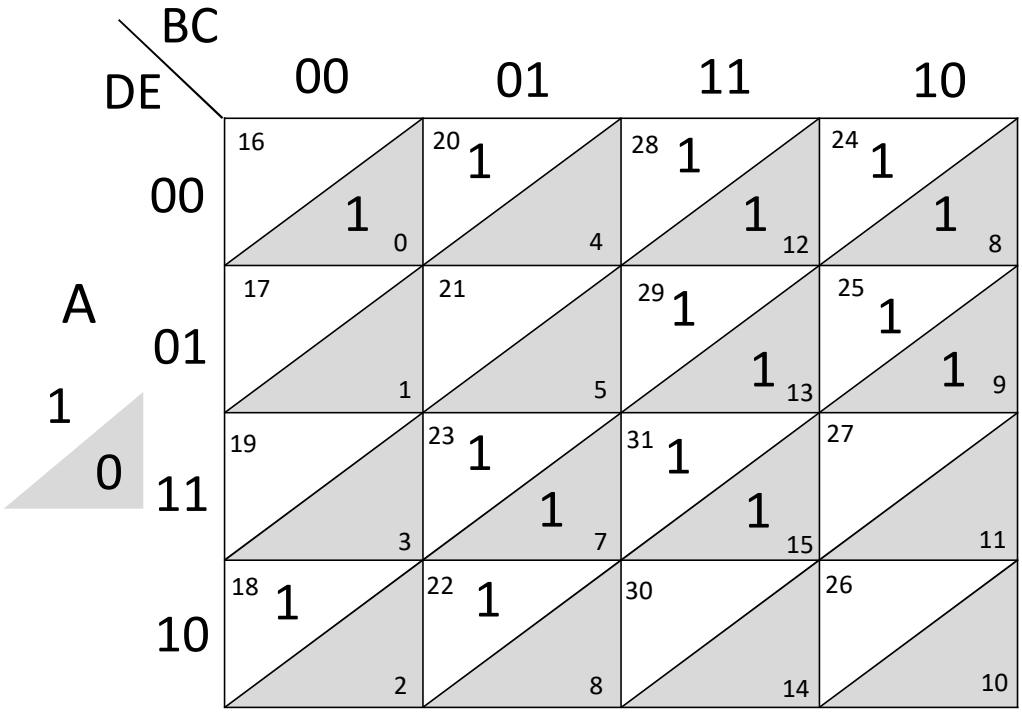
(p. 148)



陰影部分的 1 只被一個質含項所包含。

5.5 Five-variable Karnaugh Maps

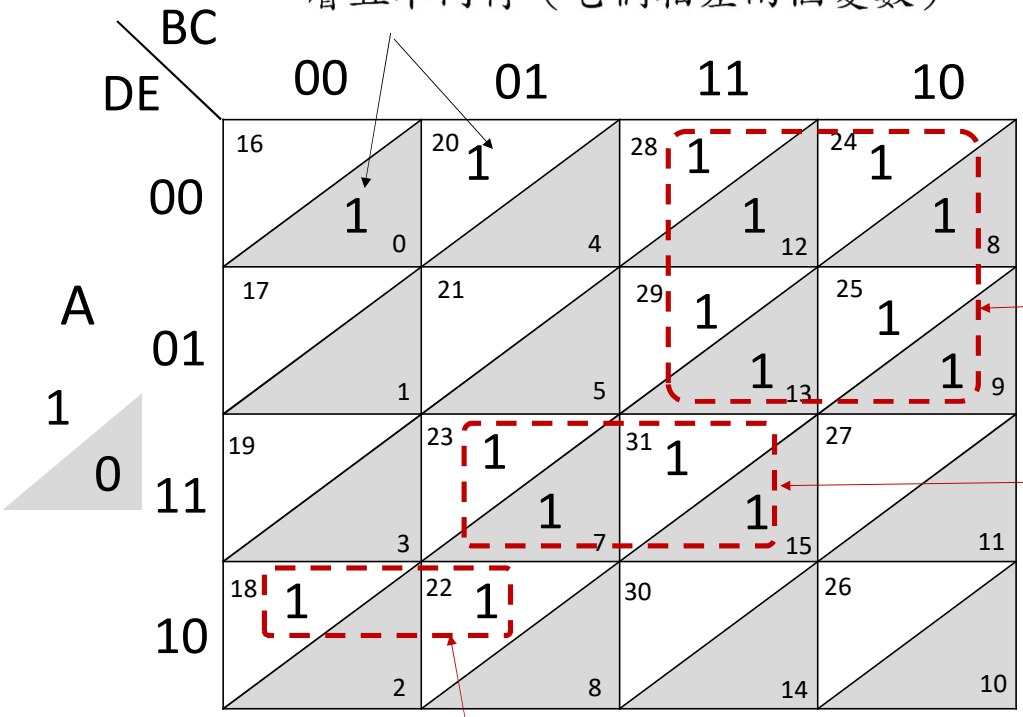
(p. 149)



(p. 149)

5.5 Five-variable Karnaugh Maps

這兩項不能合併，因為它們是在不同層且不同行（它們相差兩個變數）。



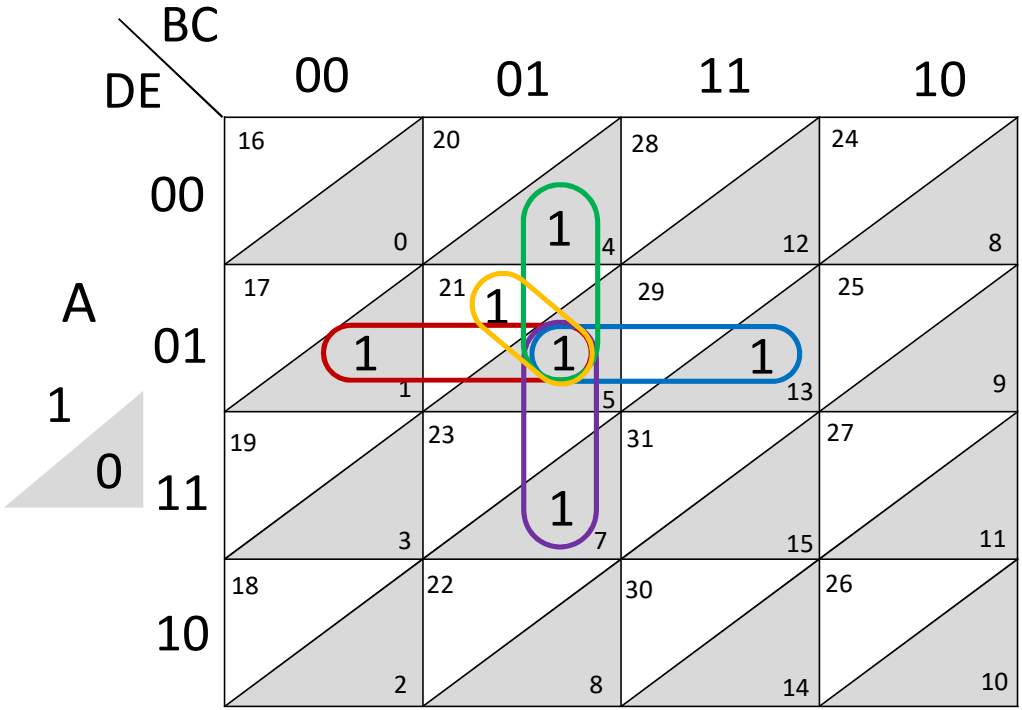
這八項合併得到BD'（B由最後兩行且D'由最上面兩列可得；因為在最上層有四項，最下層有四項，所以消去A）。

這四項（最上層兩項和最下層兩項）合併產生CDE（C由中間兩行，而DE由列可得）。

這兩項在最上層合併最到$AB'DE'$。

(p. 149)

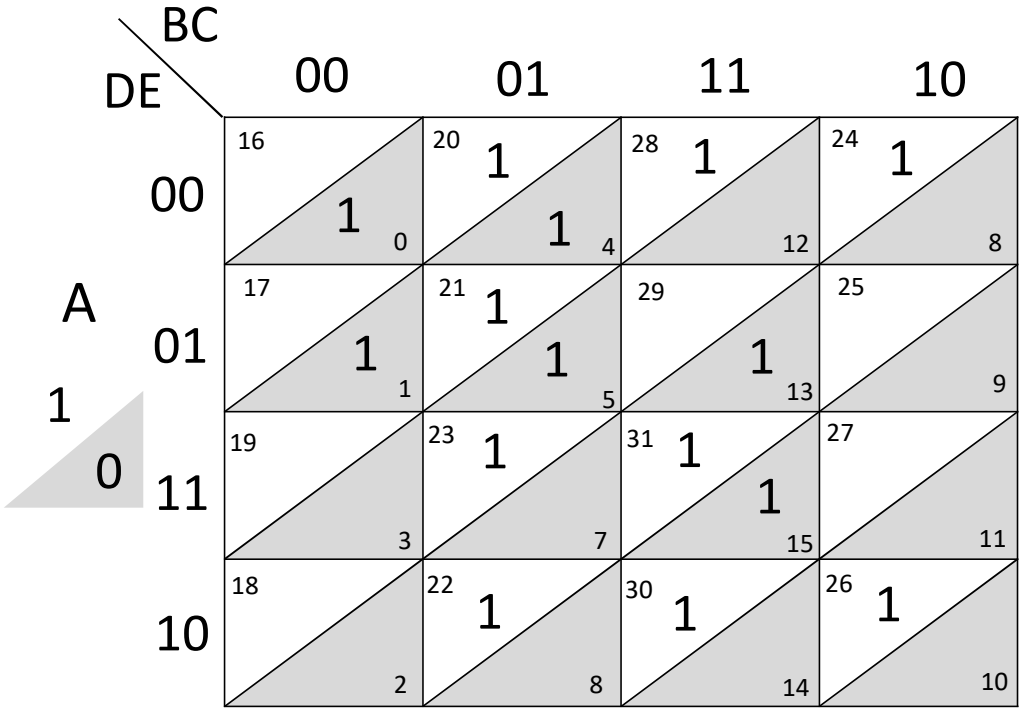
5.5 Five-variable Karnaugh Maps



$f = B'A'D'E + A'B'CD' + A'CD'E + A'B'CE' + B'CD'E$

(p. 149)

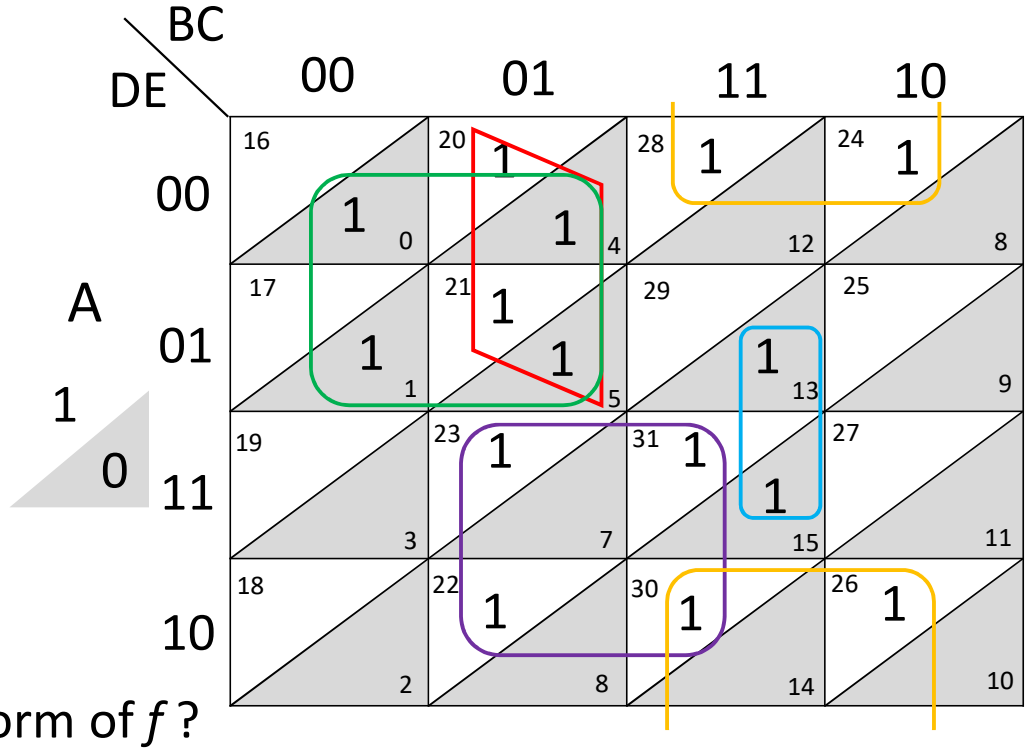
5.5 Five-variable Karnaugh Maps



$$F(A,B,C,D,E) = \sum m(0,1,4,5,13,15,20,21,22,23,24,26,28,30,31)$$

5.5 Five-variable Karnaugh Maps

(p. 149)

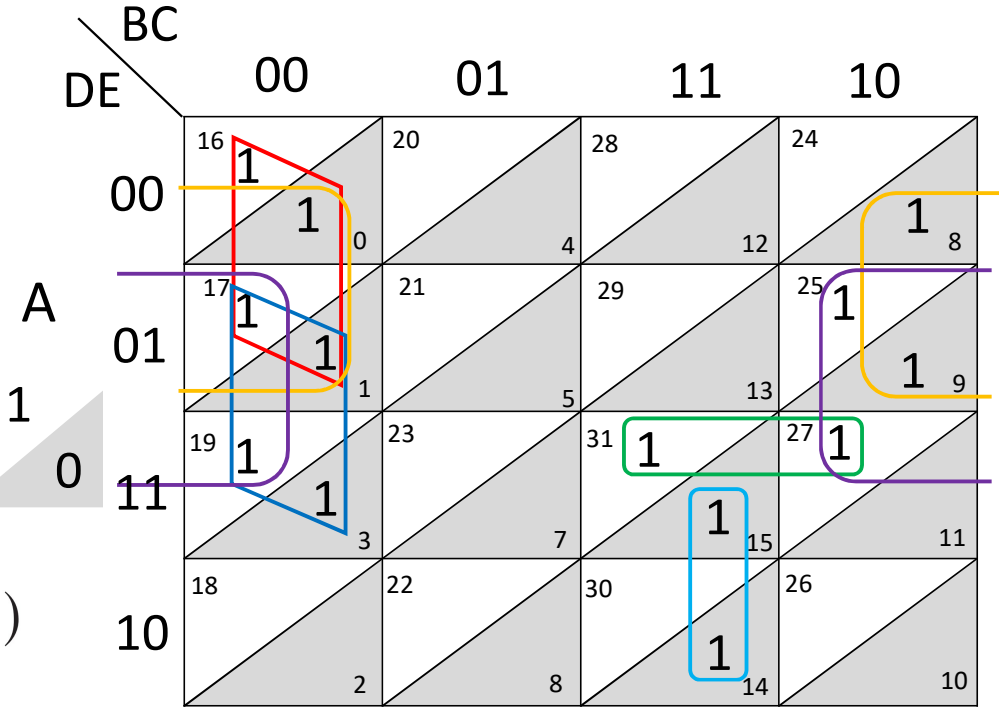


- What is the simplest form of f ?
- What are the prime implicants of f?
- What is the essential prime implicant of f ?

5.5 Five-variable Karnaugh Maps

(p. 149)

$$F(A, B, C, D, E) = \Sigma\, m(0, 1, 3, 8, 9, 14, 15, 16, 17, 19, 25, 27, 31)$$



- What is the simplest form of *f* ?
- What are the prime implicants of *f*?
- What is the essential prime implicant of *f* ?

- Skip 5.6 and 5.7, which will not be shown in the exams.
- Please practice programmed exercises by yourself from p. 154 to 159.
- Problems 5.47 to 5.50 are related to Ch. 5.6 and 5.7, which will not be included in the exams either.